

Relational Calculus

Lecture 04

Borough of Manhattan Community College

The Relational Calculus

- ❑ The relational calculus is another theoretical language that illustrates the basic operations required of any DML.
- ❑ The relational algebra (previous lecture) is a **procedural language**: a certain order is always explicitly specified in a relational algebra expression and a strategy for evaluating the query is implied.
- ❑ The relational calculus is a **nonprocedural language**: there is no description of how to evaluate a query; a relational calculus query specifies ***what*** is to be retrieved rather than ***how*** to retrieve it.

The Relational Calculus

- ❑ The relational calculus can be used to formulate the definition of a relation in terms of one or more database relations.
- ❑ The relational algebra and relational calculus are formally equivalent to one another: for every expression in the algebra, there is an equivalent expression in the calculus (and vice versa).
- ❑ The relational calculus takes its name from a branch of symbolic logic called **predicate calculus**.

The Relational Calculus

- ❑ A predicate is a truth-valued function with arguments.
- ❑ When we substitute values for the arguments, the function yields an expression, called a proposition, which can be either true or false.
- ❑ For example, the sentences, “John White is a member of staff” and “John White earns more than Ann Beech” are both propositions.

The Relational Calculus

- ❑ If a predicate contains a variable, as in “ x is a member of staff,” there must be an associated **range** for x .
- ❑ When we substitute some values of this range for x , the proposition may be true; and for other values, it may be false.
- ❑ If P is a predicate, then we can write the set of all x such that P is true for x , as: $\{x \mid P(x)\}$.
- ❑ We may connect predicates by the logical connectives (*AND*), (*OR*), and $\sim$ (*NOT*) to form compound predicates.

The Relational Calculus

- ❑ In the relational calculus, we are interested in finding tuples for which a predicate is true.
- ❑ A **tuple variable** is a variable that “ranges over” a named relation: that is, a variable whose only permitted values are tuples of the relation.
- ❑ For example, to specify the range of a tuple variable S as the Staff relation, we write: Staff(S).
- ❑ To express the query “Find the set of all tuples S such that $F(S)$ is true,” we can write:

$$\{S \mid F(S)\}$$

F is called a formula.

The Relational Calculus

- To express the query: “Find the staffNo, fName, lName, position, sex, DOB, salary, and branchNo of all staff earning more than £10,000” we can write:

$$\{S \mid \text{Staff}(S) \wedge S.\text{salary} > 10000 \}$$

S.salary means the value of the salary attribute for the tuple variable *S*.

- To retrieve particular attributes, such as first name, last name and salary, we would write:

$$\{S.\text{fName}, S.\text{lName}, S.\text{salary} \mid \text{Staff}(S) \wedge S.\text{salary} > 10000 \}$$

The existential and universal quantifiers

- There are two **quantifiers** we can use with formulae to tell how many instances the predicate applies to:
- The **existential quantifier** $\exists$ (“there exists”) is used in formulae that must be true for at least one instance, such as:

$$(\exists B) (\text{Branch}(B) \text{ B.city} = \text{'London'})$$

This means, "There exists a Branch tuple that is located in London".

- The **universal quantifier** $\forall$ (“for all”) is used in statements about every instance, such as:

$$(\forall B) (\text{Branch}(B) \text{ B.city} \neq \text{'Miami'})$$

This means, “For all Branch tuples, the address is not in Miami”.

The existential and universal quantifiers

- We can apply a generalization of De Morgan's laws to the existential and universal quantifiers:

- $(\exists X) (F(X)) \equiv \sim(\forall X) (\sim(F(X)))$

- $(\forall X) (F(X)) \equiv \sim(\exists X) (\sim(F(X)))$

- Using these equivalence rules, we can rewrite the formula:

$$(\forall B) (\text{Branch}(B) \rightarrow B.\text{city} \neq \text{'Miami'})$$

as:

$$\sim(\exists B) (\text{Branch}(B) \rightarrow B.\text{city} = \text{'Miami'})$$

which means, “There are no branches with an address in Miami”.

Expressions and Formulas

- An expression in the tuple relational calculus has the following general form:

$$\{S_1.a_1, S_2.a_2, \dots, S_n.a_n \mid F(S_1, S_2, \dots, S_m)\} \quad m \geq n$$

where $S_1, S_2, \dots, S_n, \dots, S_m$ are tuple variables; each a_i is an attribute of the relation over which S_i ranges; and F is a formula.

- A formula is made out of one or more *atoms*, where an atom has one of the following forms:

- $R(S_i)$, where S_i is a tuple variable and R is a relation.
- $S_i.a_1 \theta S_j.a_2$, where S_i and S_j are tuple variables, a_1 is an attribute of the relation over which S_i ranges, a_2 is an attribute of the relation over which S_j ranges, and θ is one of the comparison operators ($<$, $\leq$, $>$, $\geq$, $=$, $\neq$); the attributes a_1 and a_2 must have domains whose members can be compared by θ .
- $S_i.a_1 \theta c$, where S_i is a tuple variable, a_1 is an attribute of the relation over which S_i ranges, c is a constant from the domain of attribute a_1 , and θ is one of the comparison operators.

The Relational Calculus

- *List the names of all managers who earn more than £25,000.*

$\{S.fName, S.lName \mid Staff(S) \wedge S.position = 'Manager' \wedge S.salary > 25000\}$

- *List the staff who manage properties for rent in Glasgow.*

$\{S \mid Staff(S) \wedge (\exists P) (PropertyForRent(P) \wedge (P.staffNo = S.staffNo) \wedge P.city = 'Glasgow')\}$

The Relational Calculus

- List the staff who manage properties for rent in Glasgow.

$\{S \mid \text{Staff}(S) \ (\exists P) (\text{PropertyForRent}(P) \ (P.\text{staffNo} = S.\text{staffNo}) \ P.\text{city} = \text{'Glasgow'})\}$

Note that in this formulation there is no indication of a strategy for executing the query. The DBMS is free to decide the operations required to fulfill the request and the execution order of these operations.

- On the other hand, the equivalent *relational algebra* formulation would be: "Select tuples from **PropertyForRent** such that the city is **Glasgow** and perform their join with the **Staff** relation," which has an implied order of execution.

$$Staff \bowtie \Pi_{staffNo} (\sigma_{city='Glasgow'} (PropertyForRent))$$

The Relational Calculus

- *List the names of staff who currently do not manage any properties.*

$\{S.fName, S.lName \mid Staff(S) \sim (\exists P) (PropertyForRent(P) \wedge (S.staffNo = P.staffNo))\}$

- *List the names of clients who have viewed a property for rent in Glasgow.*

$\{C.fName, C.lName \mid Client(C) \wedge (\exists V) (\exists P) (Viewing(V) \wedge PropertyForRent(P) \wedge (C.clientNo = V.clientNo) \wedge (V.propertyNo = P.propertyNo) \wedge P.city = 'Glasgow')\}$

Client (clientNo, fName, lName, telNo, prefType, maxRent, eMail)

Viewing (clientNo, propertyNo, viewDate, comment)

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo, staffNo, branchNo)

❑ List the names of clients who have viewed a property for rent in Glasgow.

$\{C.fName, C.lName \mid Client(C) \quad (\exists V) (\exists P) (Viewing(V) \quad PropertyForRent(P) \\ (C.clientNo = V.clientNo) \quad (V.propertyNo = P.propertyNo) \quad P.city = 'Glasgow')\}$

Client

clientNo	fName	lName	telNo	prefType	maxRent	eMail
CR76	John	Kay	0207-774-5632	Flat	425	john.kay@gmail.com
CR56	Aline	Stewart	0141-848-1825	Flat	350	astewart@hotmail.com
CR74	Mike	Ritchie	01475-392178	House	750	mritchie01@yahoo.co.uk
CR62	Mary	Tregear	01224-196720	Flat	600	maryt@hotmail.co.uk

Viewing

clientNo	propertyNo	viewDate	comment
CR56	PA14	24-May-13	too small
CR76	PG4	20-Apr-13	too remote
CR56	PG4	26-May-13	
CR62	PA14	14-May-13	no dining room
CR56	PG36	28-Apr-13	

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

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- *List all the cities where there is a branch office but no properties for rent.*

$\{B.city \mid \text{Branch}(B) \wedge \neg(\exists P) (\text{PropertyForRent}(P) \wedge (B.city = P.city))\}$

- *List all cities where there is both branch offices and properties for rent.*

$\{B.city \mid \text{Branch}(B) \wedge (\exists P) (\text{PropertyForRent}(P) \wedge (B.city = P.city))\}$



Examples of Relational Calculus Queries

The Relational Calculus - Practice

Library (libraryNo, address, city, zipCode)

Book (bookNo, title, author, year, price, available, libraryNo)

User (userNo, name, age)

RentBook (bookNo, dateOut, dateDue, userNo)

□ formulate the following queries in the **relational calculus**:

1. List all book titles.
2. List details of all available books.
3. List the name and age of all users under age 18.
4. List the title and the year of all books published after 1960.
5. List all the books from New York libraries.
6. List the books that have never been rented.
7. List the users who rented a book written by the author 'Agatha Christie'.

Library	(<u>libraryNo</u> , address, city, zipCode)
Book	(<u>bookNo</u> , title, author, year, price, available, libraryNo)
User	(<u>userNo</u> , name, age)
RentBook	(<u>bookNo</u> , <u>dateOut</u> , dateDue, userNo)

1. List all book titles.
{ **B.title** | **Book(B)** }
2. List details of all available books.
{ **B** | **Book(B)** **B.available** = 'Y' }
3. List the name and age of all users under age 18.
{ **U.name**, **U.age** | **User(U)** **U.age** < 18 }
4. List the title and the year of all books published after 1960.
{ **B.title**, **B.year** | **Book(B)** **B.year** > 1960 }

Library	(<u>libraryNo</u> , address, city, zipCode)
Book	(<u>bookNo</u> , title, author, year, price, available, libraryNo)
User	(<u>userNo</u> , name, age)
RentBook	(<u>bookNo</u> , <u>dateOut</u> , dateDue, userNo)

5. List all the books from New York libraries.

$\{ B \mid \text{Book}(B) (\exists L) (\text{Library}(L) (L.\text{libraryNo} = B.\text{libraryNo}) (L.\text{city} = \text{'New York'})) \}$

6. List the books that have never been rented.

$\{ B \mid \text{Book}(B) \sim(\exists R) (\text{RentBook}(R) (R.\text{bookNo} = B.\text{bookNo})) \}$

Or $\{ B \mid \text{Book}(B) (\forall R) (\text{RentBook}(R) (R.\text{bookNo} \neq B.\text{bookNo})) \}$

7. List the users who rented a book written by the author 'Agatha Christie'.

$\{ U \mid \text{User}(U) (\exists R)(\exists B) (\text{RentBook}(R) \text{Book}(B) (U.\text{userNo} = R.\text{userNo}) (R.\text{bookNo} = B.\text{bookNo}) (B.\text{author} = \text{'Agatha Christie'})) \}$